AUTOMATIC HOT WATER TREATMENT SYSTEM FOR BANANA SUCKERS

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OBJECTIVES

- To eliminate soil-borne pests, nematodes, and fungal pathogens that could otherwise hinder growth or cause diseases in banana plants.
- To maintain a constant temperature for 30 min, to eliminate pests in banana suckers.
- To reducing the need for chemical treatments, contributing to sustainable and eco-friendly banana cultivation practices.
- To provide proper treatment for banana suckers, and automated system to monitor and control the temperature and time.

ABSTRACT

This project involves developing a system that heats water to a consistent 52°C, maintained by a heater controlled through a relay module, while a temperature sensor continuously monitors the water.Once the water reaches the set temperature, the system holds it steady for 30 minutes. After this time, an alert buzzer sounds, and the power to the heater is cut off automatically

INTRODUCTION

- The agricultural industry faces challenges in protecting crops from diseases and pests, especially in horticulture where plants are susceptible to infestations that can impact crop quality and yield.

- The "Automated Hot Water System for Banana Suckers" project aims to address these challenges by providing a fully automated solution for treating banana suckers through temperature-controlled water immersion.

- This automated system enhances efficiency, accuracy, and safety in agricultural treatment processes. By minimizing human involvement, it reduces labor costs and increases reliability, offering a scalable and cost-effective solution for farmers.

PROBLEM IDENTIFICATION

- Before being planted, banana suckers, or immature banana plants, are traditionally submerged in hot water to get rid of pests and diseases.
- This is guarantee that the water reaches and maintains the necessary temperature, the traditional approach necessitates continuous manual monitoring, which is a labor-intensive, time-consuming, and inconsistent procedure
- During the treatment, an operator is needed to monitor the temperature and timing, manually adjusting heat as needed to maintain the temperature.
- Manually maintaining water at a precise temperature of 52°C is difficult, especially over an extended period. Even slight deviations can reduce treatment effectiveness or damage the banana suckers, impacting plant health.

LITERATURE REVIEW

S.No	Title	Author & Year	Journal/Reference Name	Remarks
1.	Enhancing Banana Crop Viability through Temperature-Controlled Treatments	Cheng L & 2022	Journal of Horticultural Science and Biotechnology	This study highlights the potential for using organic amendments in conjunction with temperature control to achieve a sustainable approach to pest management.
2.	Effects of Temperature-Based Pest Control on Banana Suckers	Jones M & 2020	Agricultural Research Journal	They found this method to be effective in reducing both nematode and weevil infestations, which are major threats in banana cultivation.
3.	Hot Water and Organic Amendments for Nematode Control in Banana Suckers	Kumar, V& 2021	Journal of Plant Protection Research,	This study underscores the effectiveness of sustainable pest control measures that integrate thermal and organic strategies, indicating that automated temperature monitoring and control systems could optimize these treatments

LITERATURE REVIEW

S.No	Title	Author & Year	Journal/Reference Name	Remarks
4.	Optimizing Temperature-Based Treatments for Nematode Suppression in Banana Cultivation	Patel R & 2019	International Journal of Agricultural Research	They concluded that such treatments, when coupled with automated systems, could provide consistent pest management, reducing the need for chemical pesticides
5.	Efficacy of Hot Water Treatment for Nematode Control in Banana Sucker	Smith, A & 2018	International Journal of Plant Protection	Their findings showed that this method successfully reduced nematode infestations without compromising plant health.

EXISTING SYSTEM

Traditional methods involve manually heating water to 52°C in large containers and dipping the banana suckers for 20-30 minutes. This process requires constant monitoring and manual temperature control, which can lead to inconsistency in treatment quality and requires significant labor.



EXISTING SYSTEM

EXISTING PROCESS

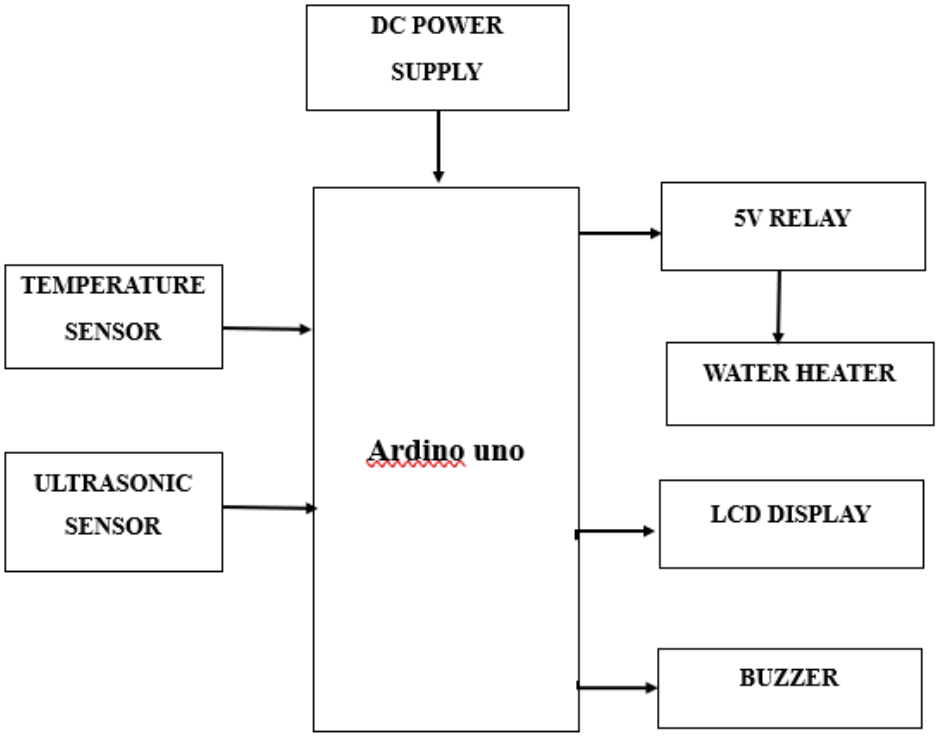
- Water Heating
- Manual Temperature Monitoring
- Dipping of Banana Suckers
- Constant Supervision



PROPOSED SYSTEM

- Maintains water at 52°C using a temperature sensor and relay, ensuring precise and consistent treatment.
- Automatically monitors and manages the 30-minute duration after reaching the target temperature.
- Displays temperature and time updates on an LCD for user convenience.
- Signals the end of the treatment cycle and automatically turns off the heater for safety.

PROPOSED SYSTEM



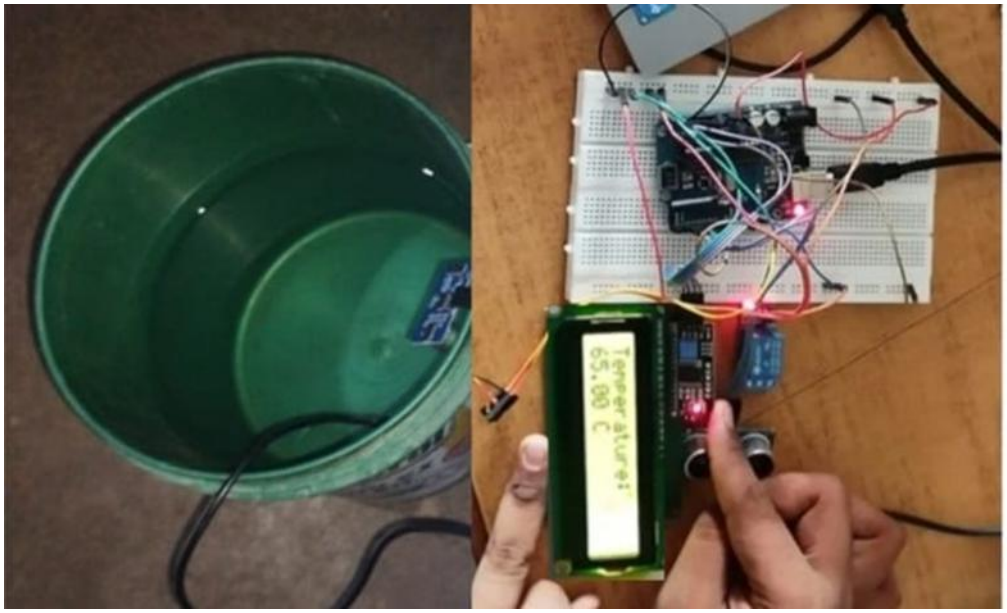
HARDWARE REQUIREMENTS

- Arduino UNO
- Ultrasonic Sensor
- DS18B20 Temperature Sensor
- Relay
- Heater
- LCD Display
- I2C Convertor
- Buzzer

SOFTWARE REQUIREMENTS

- Arduino IDE
  - Embedded C

RESULTS AND DISCUSSION



The hardware implementation of the automated hot water system for banana suckers utilizes an Arduino Uno as the central controller. . The system integrates several key components to automate temperature regulation, monitor water level, and track the treatment cycle.

ADVANTAGES & APPLICATIONS

Advantages

- The automated hot water treatment system for banana suckers that enhances precision, efficiency, and consistency in pest and disease management
- Minimizes manual monitoring, allowing more efficient use of labor and reducing human error.
- Automatic water level checks and temperature cutoff protect the system from overheating or running dry.
- LCD display and buzzer provide instant feedback on temperature and treatment completion, ensuring ease of monitoring.

ADVANTAGES & APPLICATIONS

Applications

- Suitable for both small farms and larger agricultural setups that need efficient, reliable, and easy-to-monitor plant treatment processes.
- Enables researchers to study the effects of hot water treatment on various plant diseases, contributing to better pest management techniques.

CONCLUSION

- The automated hot water system for banana suckers showcases a sophisticated integration of various technological components to optimize plant growth conditions.

- Through the integration of sensors and a relay-controlled heater, the system automatically monitors water levels, maintains the desired temperature, and provides an alert upon completion.

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THANK YOU